

Team Details



OccluSynth

Occlusion-Aware 3D Scene Reconstruction in Partially Observable Real-World Environments

Problem Statement **09**

Team

onemore_adi

Member

Aditya Agarwal

Institute

National Institute of Technology, Rourkela

Team

onemore_adi

github.com/onemore-adi/OccluSynth

Problem Statement

Selected Problem

Occlusion-Aware 3D Scene Reconstruction in Partially Observable Real-World Environments

Problem Understanding

Indoor robots see only what is in line of sight. Most of the geometry they need for safe motion is hidden behind furniture.

Classical fusion forces a binary choice: call the unseen space **free** (silent collisions) or **blocked** (paralysed robot). Neither is honest.

We reconstruct an explicit third state: **occluded but probably occupied** — recovered in 3D from RGB depth + sparse metric anchors.

The voxel states OccluSynth labels



Separating OCCLUDED (recoverable) from UNOBSERVABLE (no evidence) is what stops the system inventing geometry it cannot justify.

Proposed Solution

OccluSynth — a 5-stage pipeline that reconstructs visible geometry, completes occluded regions, and plans a collision-safe path. Every stage below is built and tested this phase.

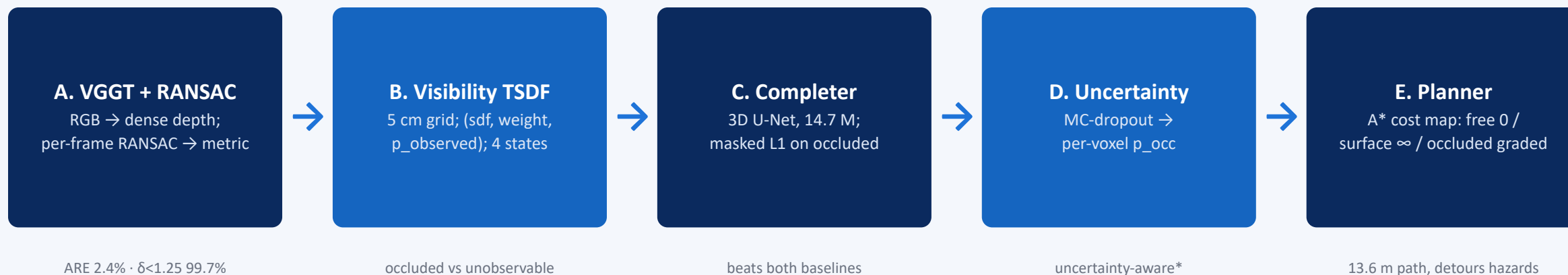


INPUT RGB frames + sparse metric anchors → **OUTPUT** dense occlusion-aware SDF + collision-safe trajectory

Every voxel reports whether it was seen or imagined — the planner uses this honesty as a soft cost, not a hard wall.

Semantics and a diffusion completer are scoped as Phase 2 — see roadmap.

Architecture



Voxel states



green = free · red = surface · amber = occluded (imagined) · grey = no evidence

System constraints & assumptions

- Poses: ScanNet GT (BundleFusion) — VGGT ATE ~70 cm is unusable at 5 cm voxels; a VIO front-end replaces this in deployment.
- Grid: 5 cm voxels · 6 frames/inference (MPS memory) · GT depth path verified.
- Cached VGGT predictions reused — fusion + completion are the interactive cost.

* uncertainty not yet calibrated — ECE 0.42; temperature scaling planned (target < 0.05).

Novelty & Innovation

The first system that tells the planner what it does not know — and measures whether it was right.

Capability	Atlas / NeuralRecon	VGGT	3D Diffusion (DiffInDScene)	Behind the Veil	OccluSynth (ours)
Reconstructs occluded geometry	×	×	○	○	✓
Works from sparse metric anchors	×	×	×	×	✓
Uncertainty-aware completion	×	○	×	×	✓
Measured on an occlusion safety benchmark	×	×	×	×	✓
Closed-loop into a navigation planner	×	×	×	×	✓

✓ yes ○ partial × no

Semantic labelling (5-class) is deferred to Phase 2 and intentionally excluded from the MVP claims above.

Results & Validation

Occluded MAE

45.3 → 27.1 cm

lower is better

Occluded F-score@5cm

0% → 32%

TSDf cannot see behind walls

Hidden-hazard awareness

0% → 21%

line-of-sight detects none

Completer — occluded voxels

Method	MAE (cm) ↓	Sign acc ↑	Compl<5cm ↑
no_completion	45.27	0.299	0.061
occluded_as_free	42.00	0.701	0.121
OccluSynth Completer	27.14	0.722	0.349

Geometry — surface vs occluded

Method	Chamfer-L1 ↓	F@5cm ↑	Occl. F ↑
TSDf-only	3.11 cm	74.1%	0.0%
OccluSynth	1.77 cm	83.5%	32.0%

Also validated: metric grounding 2.4% ARE across 10 val scenes (10/10 OK) · visibility split on scene0000_00 = 61% of observable volume occluded · planner detours around hazard clusters (15.5% collision-avoidance on scene0556_00) · 55+ test suite, 18 planner tests.

All metrics from the interim 64³ MPS checkpoint (epoch 32). The full 96³ A100 run is scripted and data-prepared — not executed this phase — and is expected to improve every number.

Build Status — what is done

Built & tested

- Metric grounding — per-frame RANSAC, ARE 2.4%, multi-scene
- Visibility-aware TSDF — 4-state grid, obliquity-corrected
- 3D U-Net completer (14.7 M) — beats both baselines
- Risk-graded A* planner — detours around hidden hazards
- MC-dropout uncertainty — per-voxel occupancy
- Occlusion safety benchmark — ScanNet-native
- Geometry eval — Chamfer-L1, F-score@5cm
- 7-Scenes cross-dataset loader — fusion runs unchanged
- 55+ automated tests across the pipeline

Partial / honest gaps

- Uncertainty not yet calibrated — ECE 0.42; temperature scaling planned (target < 0.05)
- Completer trained at interim 64³ on MPS — strong but below full resolution
- Cross-dataset probe uses 7-Scenes' own metric depth — robustness check, not a hard generalisation test
- Completer checkpoint not yet published to Hugging Face

Deferred to Phase 2

- Full 96³ A100 training run — scripted, data-prepared, awaiting compute
- 3D diffusion completer (replaces the U-Net)
- 5-class semantic head + mIoU
- Habitat-Sim navigation eval — replaced by the ScanNet-native safety benchmark
- Active perception — next-best-view from uncertainty

Open Datasets

PRIMARY

ScanNet v2

1,513 indoor RGB-D scenes with GT meshes. Drives visibility fusion, complete supervision (GT SDF from the meshes), and all geometry metrics. Non-commercial research licence.

CROSS-DATASET

7-Scenes

Microsoft tracked RGB-D sequences. A loader was implemented and the visibility fusion runs on it unchanged — a robustness/portability check (the pipeline is not ScanNet-specific).

SAFETY BENCHMARK

ScanNet-native

A hidden-hazard split: voxels that are occluded yet GT-occupied. The first reproducible occlusion-safety benchmark — and it needs no simulator (replaces the earlier Habitat-Sim plan).

Sparse metric anchors are drawn by stratified sampling ($\approx 500/\text{frame}$) to ground VGGT's scale — not uniform random.

Open Models

FROZEN BACKBONE

VGGT-Omega

Meta AI's feed-forward 3D geometry transformer (CVPR 2025 Best Paper). Used as-is, RGB → depth/pose. No fine-tuning — its geometric prior is preserved and scale is fixed downstream by RANSAC.

TRAINED (OURS)

OccluSynth Completer

3D U-Net, 14.7 M params, masked L1 on surface u occluded voxels. Predicts SDF inside the occluded volume. Checkpoint: interim_64_aug, epoch 32. HF publish pending.

PHASE 2

3D Latent Diffusion

DiffInDScene-family indoor 3D diffusion prior, to replace the U-Net completer with visibility-aware conditioning. Scoped for after the hackathon.

New work in this project: visibility-aware TSDF · RANSAC metric grounding · the completer · the risk-graded A* planner · the full evaluation harness (geometry, completer, safety, cross-dataset).

AI / GenAI in the System & Workflow

In the system

- The 3D U-Net completer is the generative component — it imagines geometry that appears in no depth image (a diffusion prior takes this over in Phase 2).
- VGGT foundation model — a strong 3D prior reused frozen, grounded to metric scale by RANSAC.
- Key insight: the visibility mask is conditioning — it tells the model exactly which voxels it is licensed to imagine, and which to leave alone.

In our workflow

- Agentic, LLM-assisted development (Claude) — architecture, ablation scripting, the evaluation harness, and the demo-video tooling.
- Literature synthesis to track 2024–25 work in 3D completion and occlusion-aware planning.
- Reproducible, test-driven harness — 55+ tests pin metrics so regressions surface immediately.

Full agentic-development writeup in docs/ax.md.

Compute, Deliverables & Roadmap

Compute

- Interim training on Apple Silicon MPS (16 GB) — no cloud GPU used.
- Full 96³ run is scripted & data-prepared: `train_completer.py --device cuda --crop_size 96`.
- Not executed this phase — compute access, not missing work (~100 A100 GPU-hr budget).

Hackathon deliverables

- Working pipeline on ScanNet scenes ✓
- Demo video — rendered & assembled from the pipeline ✓
- Quantitative comparison vs baselines (+ reported Atlas / NeuralRecon) ✓
- Open-source code + 55-test suite ✓
- Completer checkpoint published to HF — pending

Roadmap (Phase 2)

- Run the full 96³ A100 training — lifts every reported metric.
- Calibrate uncertainty — temperature scaling, target ECE < 0.05.
- Swap the U-Net for a 3D diffusion completer.
- Add the 5-class semantic head (mIoU).
- Active perception — next-best-view from uncertainty.

Top risks & mitigations

- Uncertainty miscalibration → temperature scaling / deep ensembles + ECE check.
- Low occluded-hazard recall (21%) → full-resolution A100 run + more training data.
- Pose dependence (GT poses) → VIO / SLAM front-end in real deployment.
- Cross-dataset realism → feed monocular depth instead of metric depth on 7-Scenes.

Appendix — References & Methods

State-of-the-art referenced

- Reconstruction: Atlas (ECCV'20) · NeuralRecon (CVPR'21) · TransformerFusion (NeurIPS'21) · VGGT (CVPR'25, our backbone)
- Completion / generation: DiffInDScene · Octree Latent Semantic Diffusion ('25) · Behind the Veil ('24) · RecGen ('25)
- Depth & planning: Marigold-DC · normal-guided sparse sampling · OA-MPC · RAMP (risk-aware mapping & planning)

Implementation details

- **Metric grounding** — per-frame RANSAC affine fit on ≈ 500 stratified anchors; ARE 2.4%, scale varies per scene.
- **Visibility TSDF** — projective integration, obliquity-corrected surface band; per-voxel (sdf, weight, p_observed); states free / surface / occluded / unobservable.
- **Completer** — 3D U-Net (14.7 M), masked L1 on surface $\cup$ occluded, D4 augmentation, 96^3 crops; GT SDF from meshes on the identical grid.
- **Planner** — A* on a 2D cost map at robot height; free 0 / surface ∞ / occluded graded by p_occupied.
- **Safety benchmark** — hidden hazard = occluded $\wedge$ GT-occupied; awareness + collision-avoidance vs baselines.
- **Metrics** — Chamfer-L1, F-score@5cm, sign accuracy, ECE — reported separately for surface and occluded voxels.